INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

HYDERABAD

Linguistic Data 3
Data Modeling in Indian Languages

Assignment 3

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Contents

1 Introduction **2**

2 Dataset **2**
2.1 Telugu Treebank 2
2.2 Hindi Treebank 2

3 Dependency Distance **3**

4 Implementation **3**

5 Dependency Distance Statistics **4**

6 Dependency Relations **4**

7 Morphological Feature Analysis **5**

8 Statistical Testing **5**

9 Visualization of Results **5**
9.1 Histogram Comparison 6
9.2 Boxplot Comparison 6
9.3 Density Plot 7
9.4 Python Generated Histogram 7

10 Discussion **7**

11 Conclusion **8**

1 Introduction

This assignment explores structural properties of dependency trees in Indian languages. We perform a comparative analysis between Telugu and Hindi treebanks to understand differences in syntactic structure.

Dependency distance is an important concept in computational linguistics and psycholinguistics. It measures the linear distance between a head word and its dependent in a dependency tree.

Languages with freer word order may exhibit longer dependency distances, while languages with stricter word order tend to keep related words closer together.

In this assignment we:

- Loaded and processed Telugu and Hindi treebank data.
- Computed dependency distance statistics.
- Analyzed dependency relations.
- Examined morphological feature distributions.
- Performed statistical hypothesis testing.
- Visualized the distributions using plots.

The implementation was done using Python for data processing and R for statistical analysis and visualization.

2 Dataset

Two dependency treebanks were used.

2.1 Telugu Treebank

The Telugu data comes from the IIIT-Hyderabad Telugu treebank. The file used was:

```
iiit_hcu_intra_chunk_v1.conll
```

Total tokens:

```
17262
```

2.2 Hindi Treebank

Hindi data consists of multiple CoNLL formatted files. All files were read and merged into a single dataset.

Total tokens:

```
404058
```

3 Dependency Distance

Dependency distance is defined as the absolute difference between the position of a word and the position of its head.

$$distance = |index - head|$$

Example:

```
ID  FORM  HEAD
1   this  2
2   book  0
```

Distance between "this" and its head "book" is:

$$|1 - 2| = 1$$

This metric helps quantify syntactic complexity.

4 Implementation

The analysis was implemented in Python.

Project structure:

```
assignment3

   data
       telugu
           iiit_hcu_intra_chunk_v1.conll

       hindi
           (multiple .conll files)

   src
       parser.py
       dependency_analysis.py
       morphology_analysis.py
       stats_tests.py
       main.py
       test_load.py

   results
       dependency_distance_histogram.png
       dependency_histogram.png
       dependency_boxplot.png
       dependency_density.png
       dependency_summary.csv

   README.tex
```

The Python program loads the treebank, computes statistics, and generates plots.

To run the analysis:

```
python src/main.py
```

5 Dependency Distance Statistics

Results from the Python analysis:

```
Telugu Dependency Distance Stats
Mean: 1.56
Median: 1

Hindi Dependency Distance Stats
Mean: 3.19
Median: 1
```

Observation:

Hindi shows a larger mean dependency distance than Telugu. This suggests that syntactic dependencies in Hindi tend to span longer distances.

6 Dependency Relations

Top dependency relations in Telugu:

```
rsym
root
k1
nmod
k2
vmod
adv
k7p
k7t
r6
```

Top dependency relations in Hindi:

```
lwg__psp
nmod__adj
rsym
k1
ccof
pof__cn
r6
main
k2
lwg__vaux
```

Hindi uses several complex relation labels, reflecting morphological and syntactic information encoded in the treebank.

7 Morphological Feature Analysis

Both datasets contain rich morphological annotations.

Common features include:

- cat
- gen
- num
- pers
- case
- vib
- tam
- chunkId
- chunkType
- stype

These features describe grammatical categories such as gender, number, person, and tense-aspect-mood.

8 Statistical Testing

A two-sample t-test was performed to determine whether dependency distances differ significantly between Telugu and Hindi.

```
t-statistic: -132.34
p-value: 0.0
```

Since:

$$p < 0.05$$

the difference is statistically significant.

This confirms that the two languages show different dependency distance patterns.

9 Visualization of Results

Several plots were generated to visualize the dependency distance distributions.

9.1 Histogram Comparison

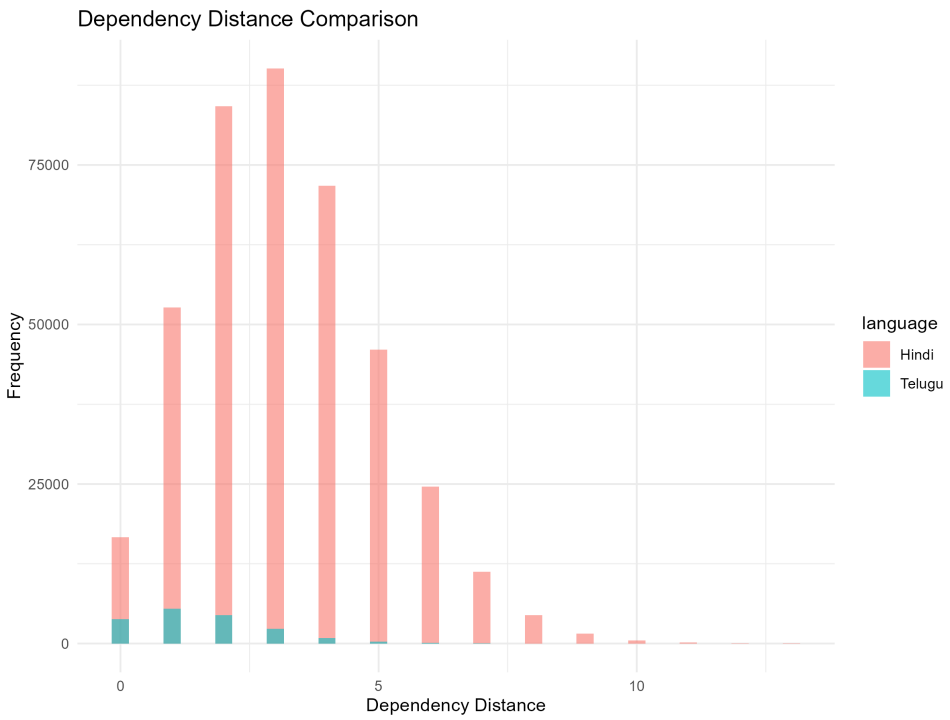

Figure 1: Histogram comparing dependency distances in Telugu and Hindi

9.2 Boxplot Comparison

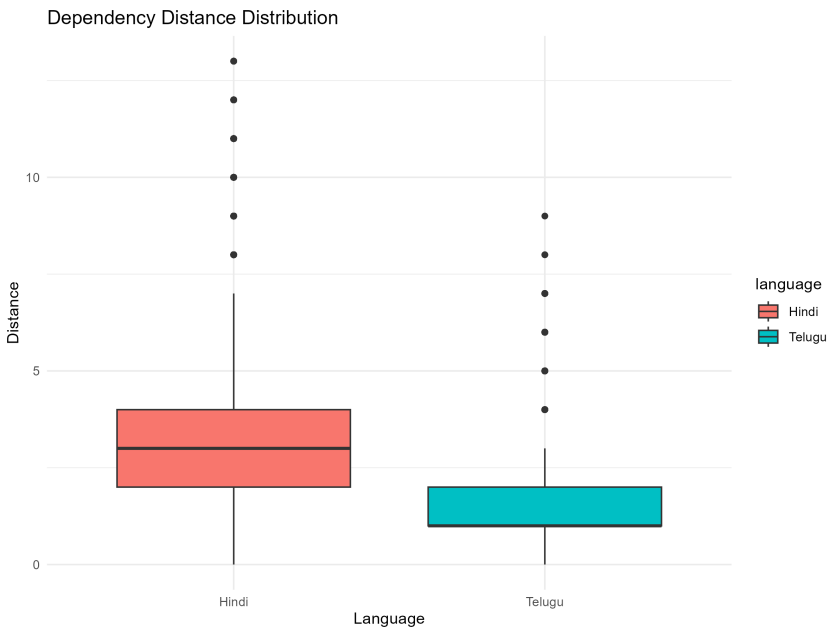

Figure 2: Boxplot showing dependency distance distributions

9.3 Density Plot

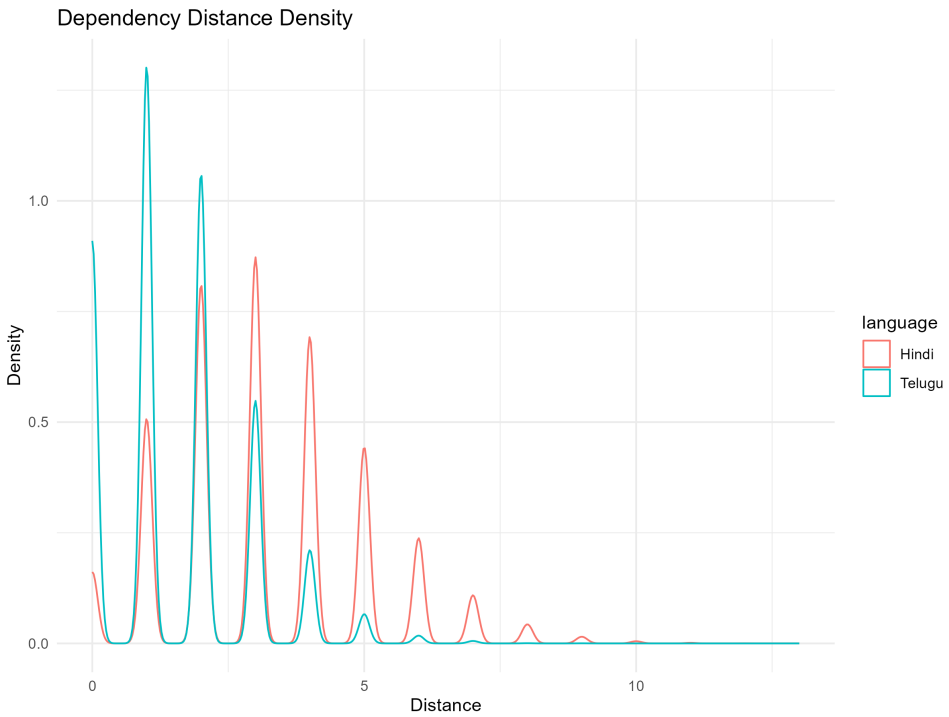

Figure 3: Density plot comparing the two languages

9.4 Python Generated Histogram

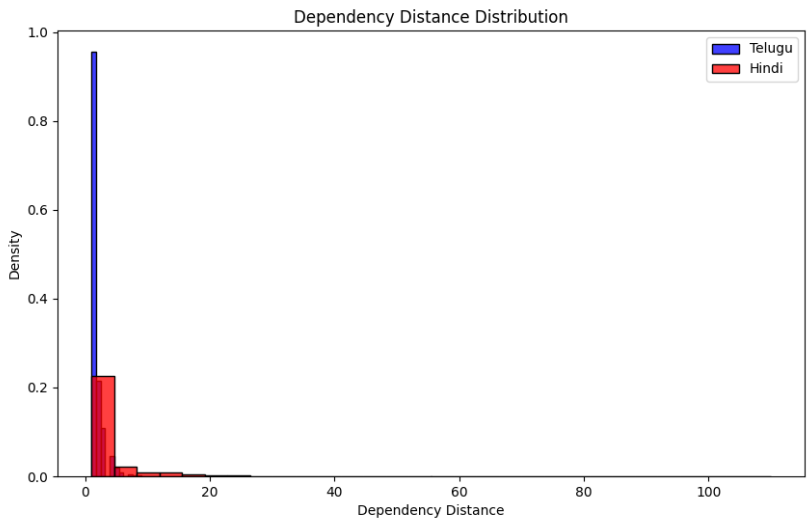

Figure 4: Dependency distance histogram generated from Python analysis

10 Discussion

The analysis reveals several interesting observations:

- Hindi exhibits larger average dependency distances compared to Telugu.
- Telugu dependencies tend to be shorter and more local.
- Morphological annotation structures are similar across the datasets.
- Dependency relations reflect the syntactic structure of each language.

These findings align with linguistic observations that Indian languages often exhibit flexible word order but still maintain strong syntactic dependencies.

11 Conclusion

In this assignment we analyzed dependency treebanks for Telugu and Hindi.

We implemented a pipeline that loads treebanks, extracts syntactic statistics, and performs comparative analysis.

Key findings include:

- Hindi has larger average dependency distances.
- The difference between the languages is statistically significant.
- Morphological annotation is extensive in both treebanks.

This type of analysis helps us understand structural properties of languages and provides insights into syntactic complexity.

Such metrics are useful in computational linguistics, psycholinguistics, and natural language processing.

Code Repository:

`https://github.com/ishitab1310/Linguistic-Data-3-Data-Modeling-in-ILs/tree/main/assignment3`